

Week 7 · Class 1

Multivariate Models & Categorical Predictors and Interactions

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Segment A — Multivariate Models

In Today's Class

Segment A — Multivariate Models

- Adding multiple predictors to regression
- “Controlling” for confounding variables
- Simpson’s paradox: when aggregation misleads
- Model building strategies
- Understanding R^2 and model fit

Segment B — Categorical Predictors and Interactions

- Including categorical variables in regression models
- Dummy variables and reference categories
- Interaction terms and conditional relationships
- Interpreting models with multiple categorical predictors

Does Trade Prevent War?

Observation: Countries that trade more with each other fight less

Simple interpretation: Trade prevents war

The problem: This ignores other important factors

- Geographic distance (neighbors trade more AND fight more)
- Economic development (rich countries trade more AND have fewer conflicts)
- Democratic institutions (democracies trade more AND fight less)

- Alliance relationships (allies trade more AND don't fight each other)

Simple correlation can be deeply misleading when confounders are present

Simpson's Paradox

A relationship that appears in aggregated data disappears or reverses when we examine sub-groups

The mechanism:

- A confounder is **unevenly distributed** across groups
- Aggregating hides that imbalance — the confounder's effect masquerades as the main relationship

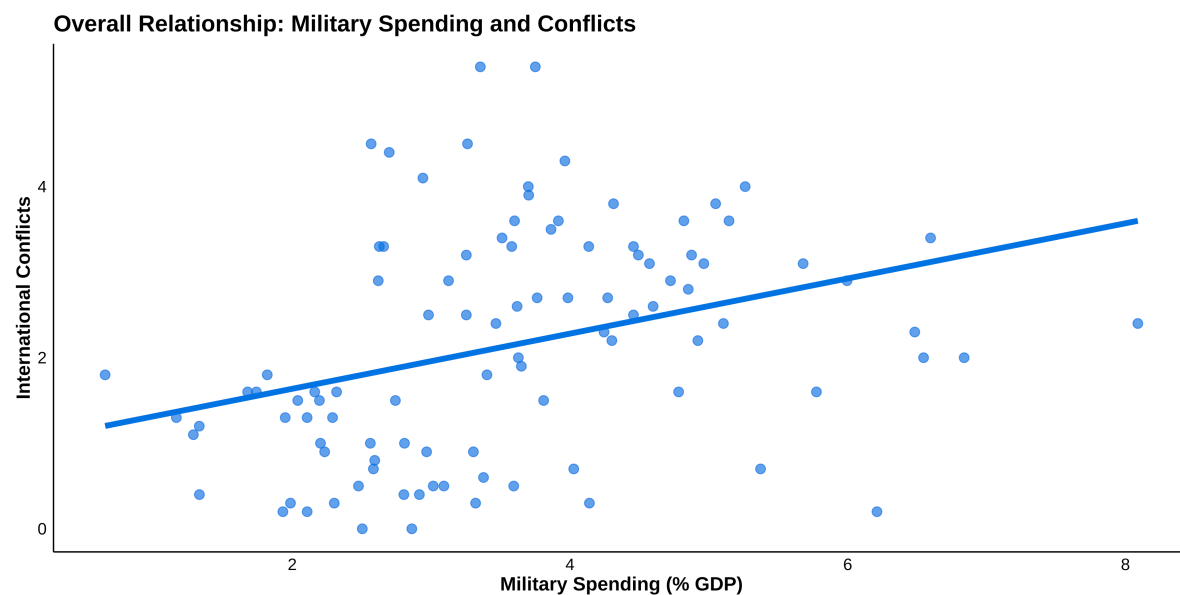
International Relations example:

- Overall: Larger military spending predicts more international conflicts
- Within regime types: Democracies with high military spending may be more peaceful
- The confounding variable (regime type) creates a misleading overall pattern

Multivariate analysis helps us see through these misleading patterns

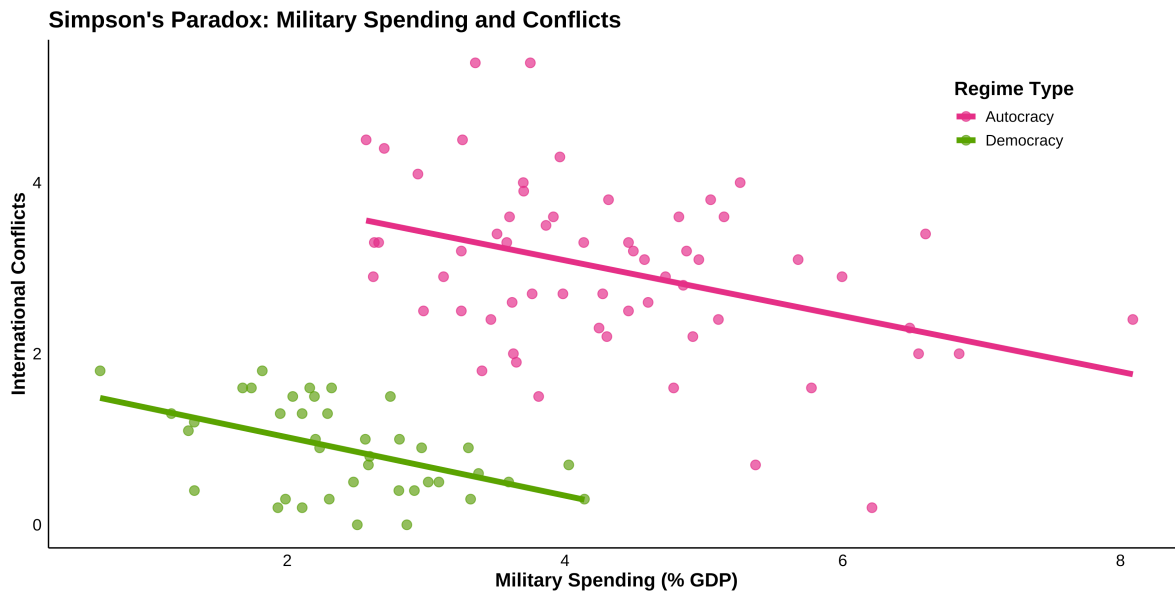
All Countries Pooled

``geom_smooth()`` using formula = 'y ~ x'



Now Break It Down by Regime Type

``geom_smooth()`` using formula = 'y ~ x'



Controlling for Confounders

What Are Confounders? (Reminder)

Confounding Variables (Z): Variables that affect both X and Y, creating spurious relationships

Key characteristics:

- Z causes both the independent variable (X) and dependent variable (Y)
- Z creates a “backdoor path” between X and Y
- Ignoring Z leads to biased estimates of X’s effect on Y
- Z explains away part of the apparent $X \rightarrow Y$ relationship

The Logic of Statistical Control

“**Holding constant**” means: compare observations that have the **same value of Z** but **differ in X**, and see how Y changes.

Example: Does education predict immigration attitudes, or is partisanship doing the work?

- **Among Democrats only:** Do college-educated Democrats support immigration more than non-college Democrats?
- **Among Republicans only:** Do college-educated Republicans support immigration more than non-college Republicans?

If the education effect is similar **within both groups**, education has a real effect beyond partisanship. Multivariate regression does this comparison across **all values of Z simultaneously**.

Eight Voters, One Question

Suppose we survey 8 people about immigration support (0–100):

Person	Education	Party	Immigration Support
1	No College	Dem	65
2	College	Dem	75
3	No College	Dem	60
4	College	Dem	70
5	No College	Rep	30
6	College	Rep	40
7	No College	Rep	25
8	College	Rep	35

The Education Gap Survives

Comparison	College Avg	No College Avg	Gap
Overall	55	45	10
Within Democrats	72.5	62.5	10
Within Republicans	37.5	27.5	10

The gap stays the same within each group → Education’s effect is real, not driven by party

If party had inflated the gap (e.g., college voters are disproportionately Democrats), the within-group gaps would **shrink**. That’s what “controlling” reveals.

When the Effect Doesn’t Survive

Now suppose we ask: does **income** predict immigration support?

Notice: most high-income people happen to be Democrats

Person	Income	Party	Immigration Support
1	High	Dem	66
2	High	Dem	64
3	High	Dem	68
4	High	Rep	32
5	Low	Dem	64
6	Low	Rep	28
7	Low	Rep	30
8	Low	Rep	32

The Effect Was Spurious

Comparison	High Income Avg	Low Income Avg	Gap
Overall	57.5	38.5	19
Within Democrats	66	64	2
Within Republicans	32	30	2

The gap shrinks from 19 to 2 within each group → Income’s apparent effect was really party doing the work. High-income people just happened to be more Democratic.

This is what “controlling” does: it separates the real effect of X from the confounding influence of Z. If the effect **survives**, X matters. If it **shrinks or disappears**, Z was the true driver.

Why Not Just Subset the Data?

Splitting by Z worked for our example — why not always do that?

Problem 1: Z is often **continuous**

If Z is age, you’d need a separate group for every value:

- Age = 31: 4 people
- Age = 32: 3 people
- Age = 33: 2 people
- Age = 34: 0 people...

You can't compute a reliable average with 2–3 people per group, and some groups are empty.

Why Not Just Subset? (continued)

Problem 2: You usually have **multiple confounders**

Suppose you need to control for party, income, age, AND region:

- Democrat + High Income + Age 35 + Northeast → **1 person**
- Republican + Low Income + Age 42 + South → **0 people**

Each additional confounder **multiplies** the number of groups. With 4 confounders you could need hundreds of groups — most will have too few observations to learn anything.

So What Does Regression Do Instead?

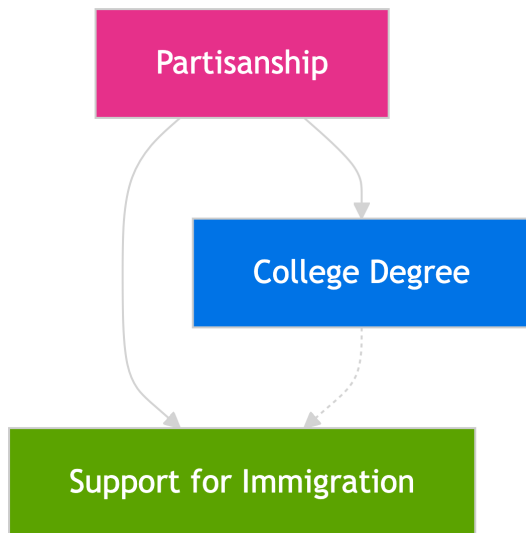
Regression is like **automatic subsetting** — it compares observations with similar Z values across the entire dataset and estimates a single coefficient for X, all in one equation.

What regression gives you:

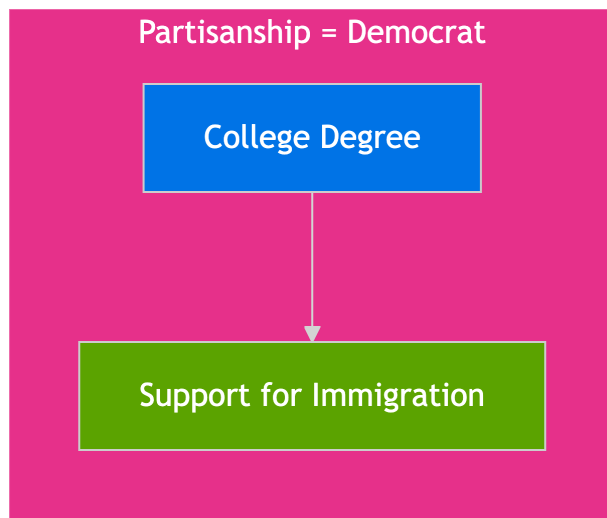
- Handles continuous and categorical confounders the same way
- Controls for multiple confounders simultaneously
- Uses **all** observations at once (no tiny subgroups)
- Produces a single, clean estimate of X's effect

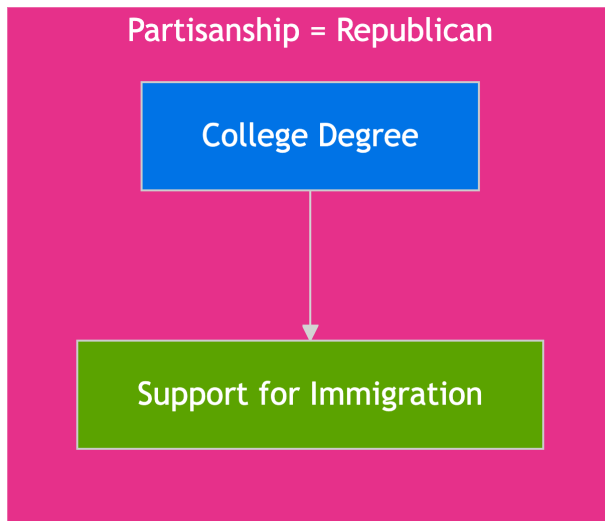
Visualizing Statistical Control

Without Control



With Control: At Fixed Z Values





Statistical control compares units with the same Z value, eliminating Z's confounding effect

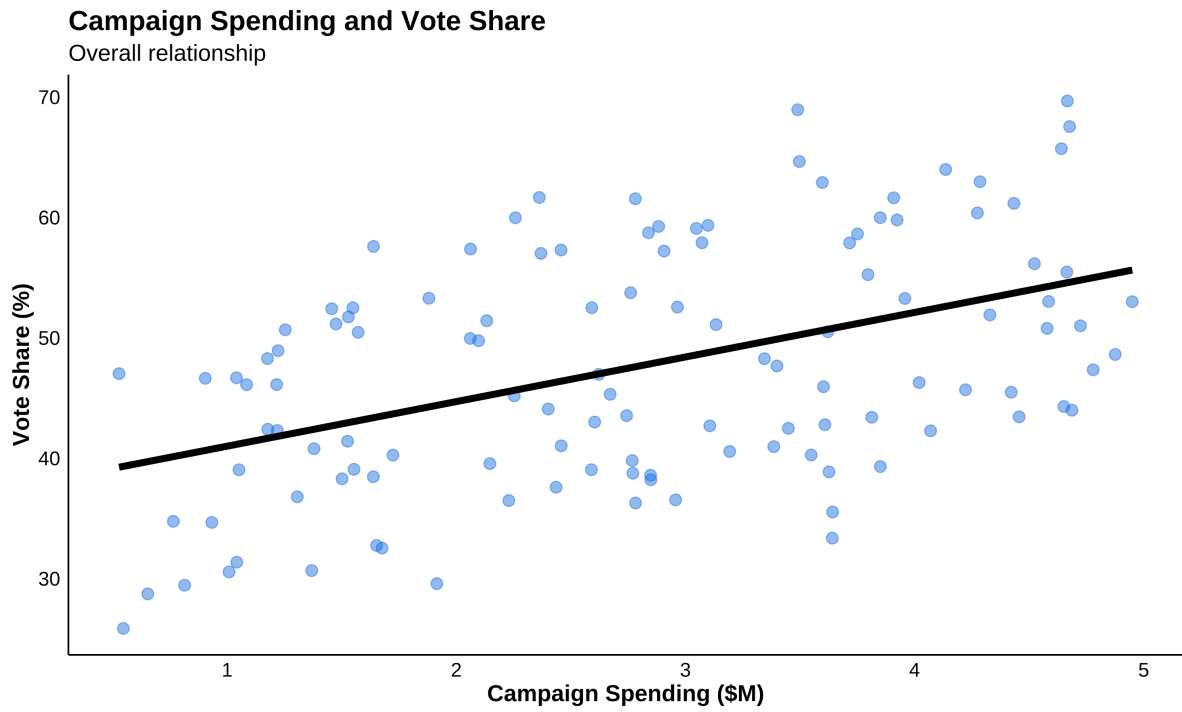
Example 1: Campaign Spending and Vote Share

Question: What's the effect of campaign spending on vote share?

Ignoring unknown labels:

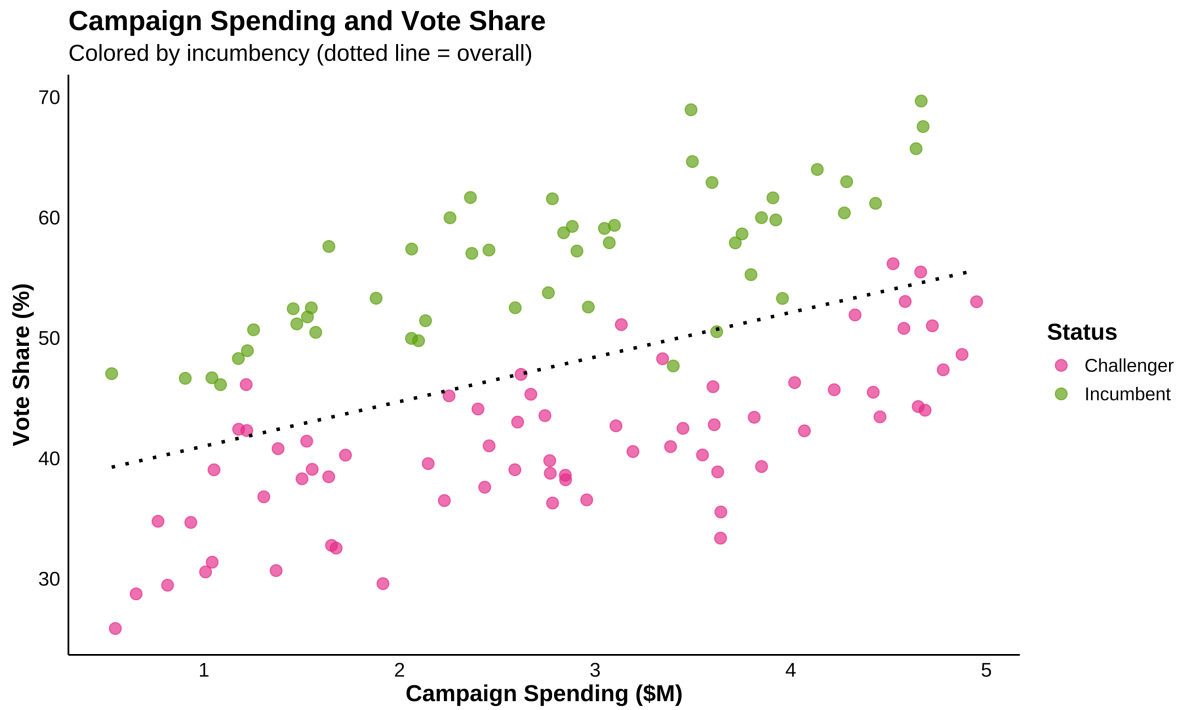
```
* colour : ""  
`geom_smooth()` using formula = 'y ~ x'
```

Warning: No shared levels found between `names(values)` of the manual scale and the data's colour values.



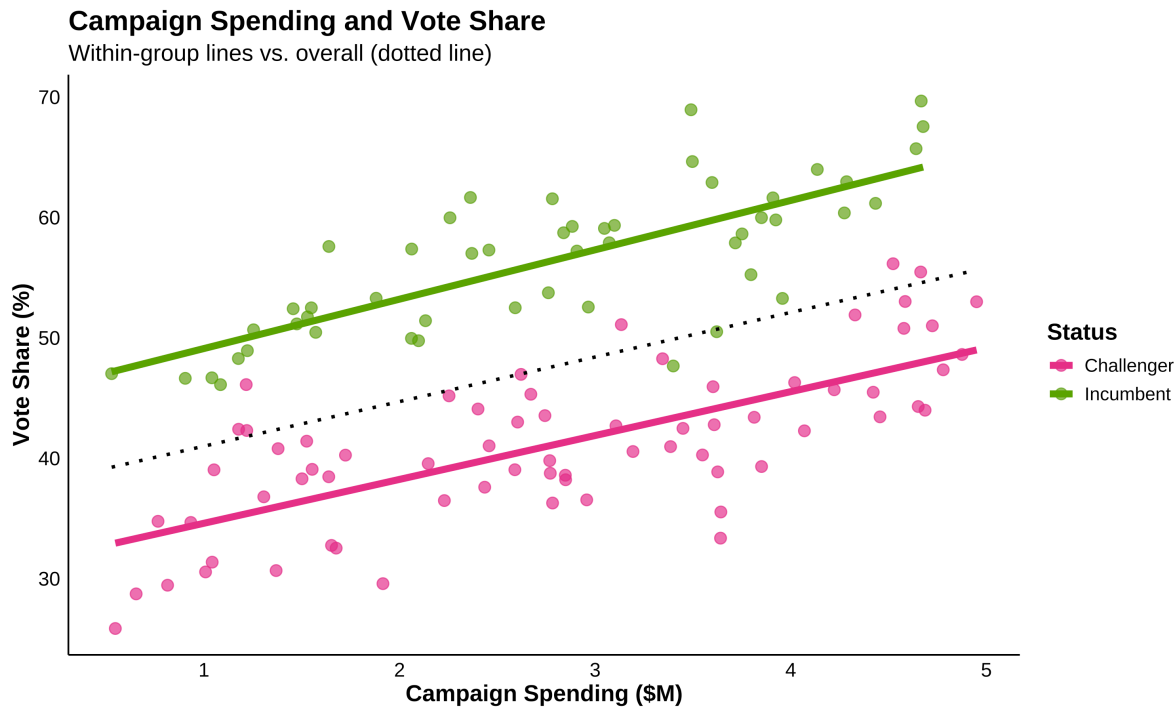
Campaign Spending and Vote Share

```
`geom_smooth()` using formula = 'y ~ x'
```



Campaign Spending and Vote Share

```
`geom_smooth()` using formula = 'y ~ x'  
`geom_smooth()` using formula = 'y ~ x'
```



Intercept shift:

- The slope barely changes — controlling for incumbency adjusts **baselines**, not the spending effect
- Incumbents just start ~15 points higher

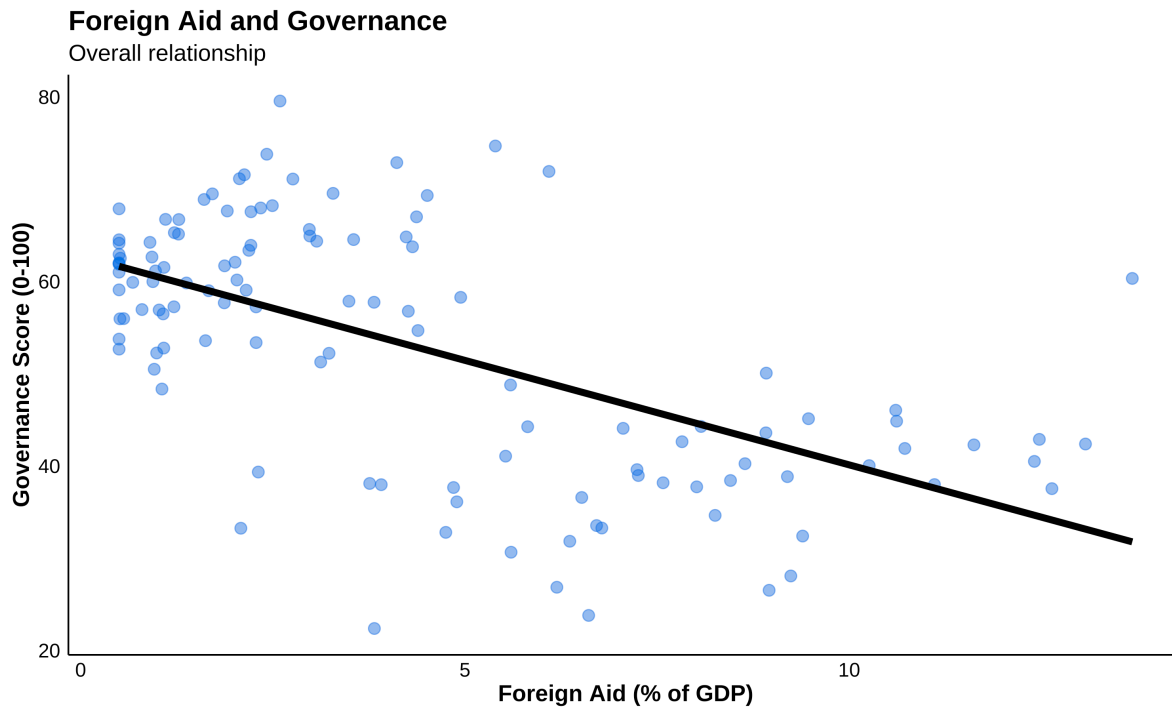
Example 2: Foreign Aid and Governance

Question: Does foreign aid improve governance?

Ignoring unknown labels:

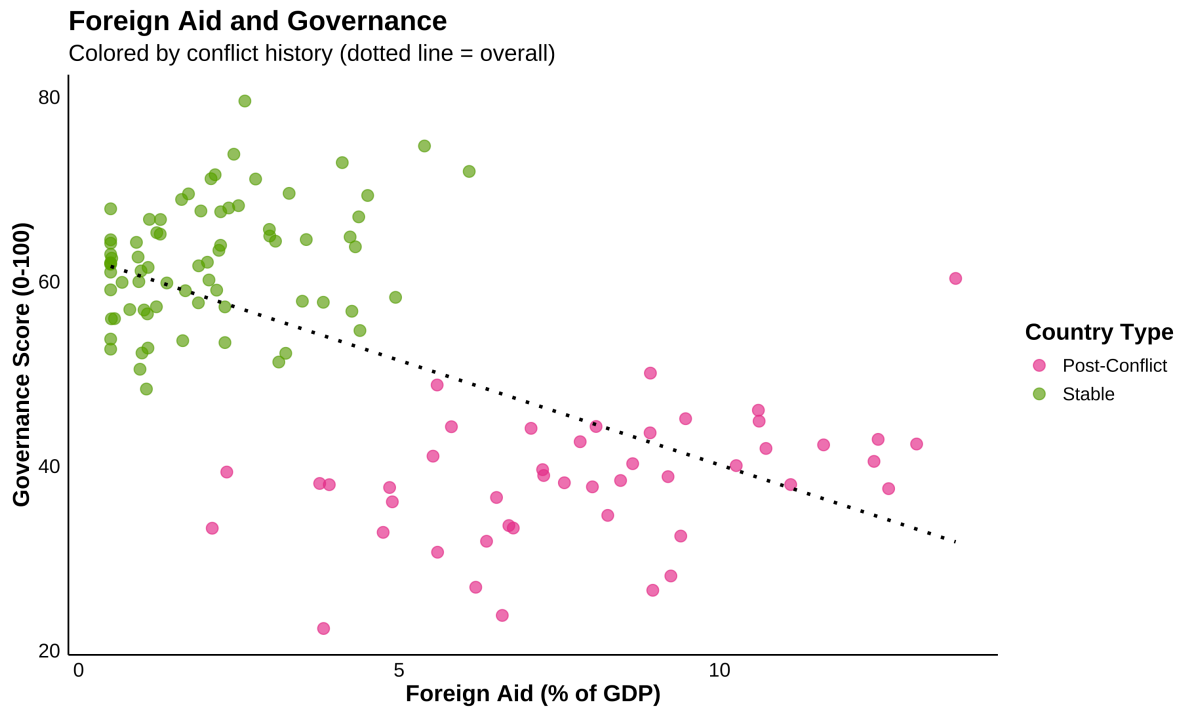
```
* colour : ""
`geom_smooth()` using formula = 'y ~ x'
```

Warning: No shared levels found between `names(values)` of the manual scale and the data's colour values.



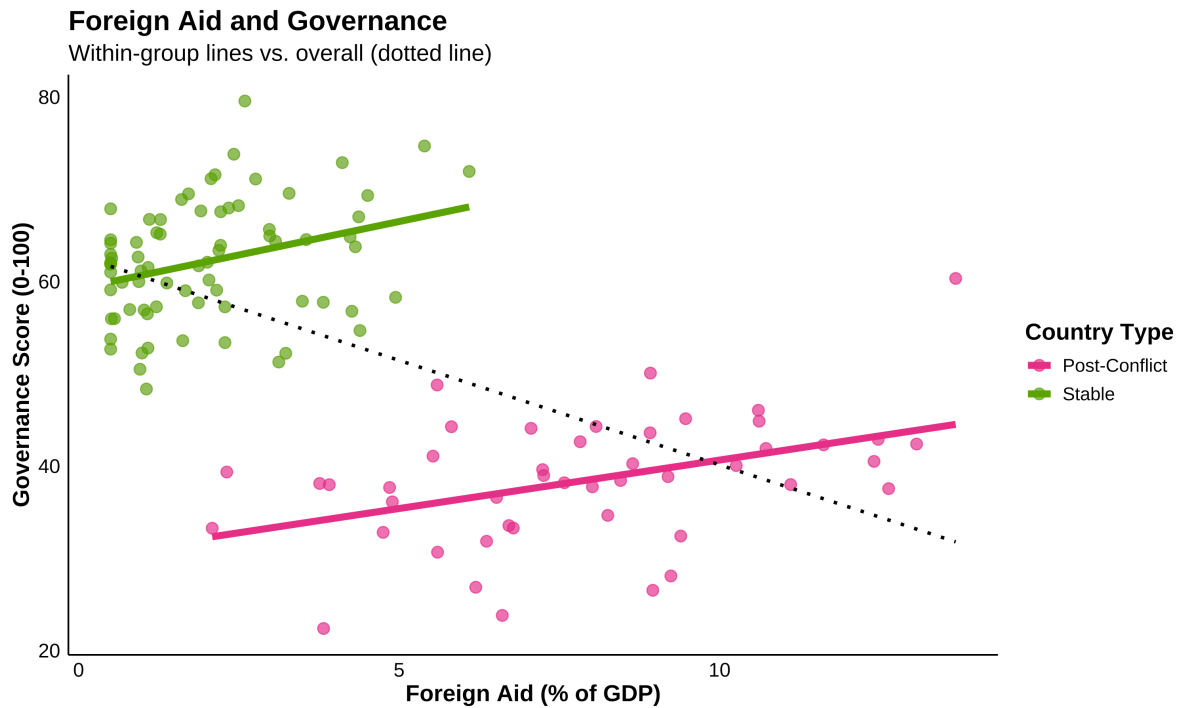
Foreign Aid and Governance

```
`geom_smooth()` using formula = 'y ~ x'
```



Foreign Aid and Governance

```
`geom_smooth()` using formula = 'y ~ x'  
`geom_smooth()` using formula = 'y ~ x'
```



A full reversal:

- The overall line slopes **down** (aid looks harmful) — but within each group, aid **helps**
- Conflict history is the confounder: post-conflict countries get more aid *and* have worse governance

Your First Multivariate Model

Question: What's the effect of age on wages for women?

- **Simple model:** Just age
- **Multivariate model:** Age and having a college degree

```
# Load motherhood wage data
motherhood_data <- read_csv("../data/motherhood.csv") %>%
  filter(!is.na(wage), !is.na(age), !is.na(educ)) %>%
  mutate(
    # Create college indicator (1 = college degree, 0 = less than college)
    college_or_more = ifelse(educ == "4.College", 1, 0),
    # Convert wage to thousands for easier interpretation
    wage_thousands = wage / 1000
  )
```

```
) %>%
filter(wage > 0, age >= 18, age <= 65)
```

New names:

Rows: 18282 Columns: 27

-- Column specification

```
----- Delimiter: "," chr
(4): educ, firmSize, marstat, industry dbl (21): ...1, PUBID, year, weight,
selfemp, wage, age, numChildren, experi... lgl (2): school, fullTime
i Use `spec()` to retrieve the full column specification for this data. i
Specify the column types or set `show_col_types = FALSE` to quiet this message.
* `` -> `...1`
```

```
# Simple model: just age
```

```
simple_model <- lm(wage_thousands ~ age, data = motherhood_data)
```

```
# Multivariate model: age + education
```

```
multi_model <- lm(wage_thousands ~ age + college_or_more, data = motherhood_data)
```

Comparing Simple vs Multivariate Results

```
# Compare the models
```

```
tidy(simple_model)
```

```
# A tibble: 2 x 5
```

	term	estimate	std.error	statistic	p.value
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	(Intercept)	-1.11	0.129	-8.63	6.96e-18
2	age	0.0969	0.00530	18.3	6.95e-74

```
tidy(multi_model)
```

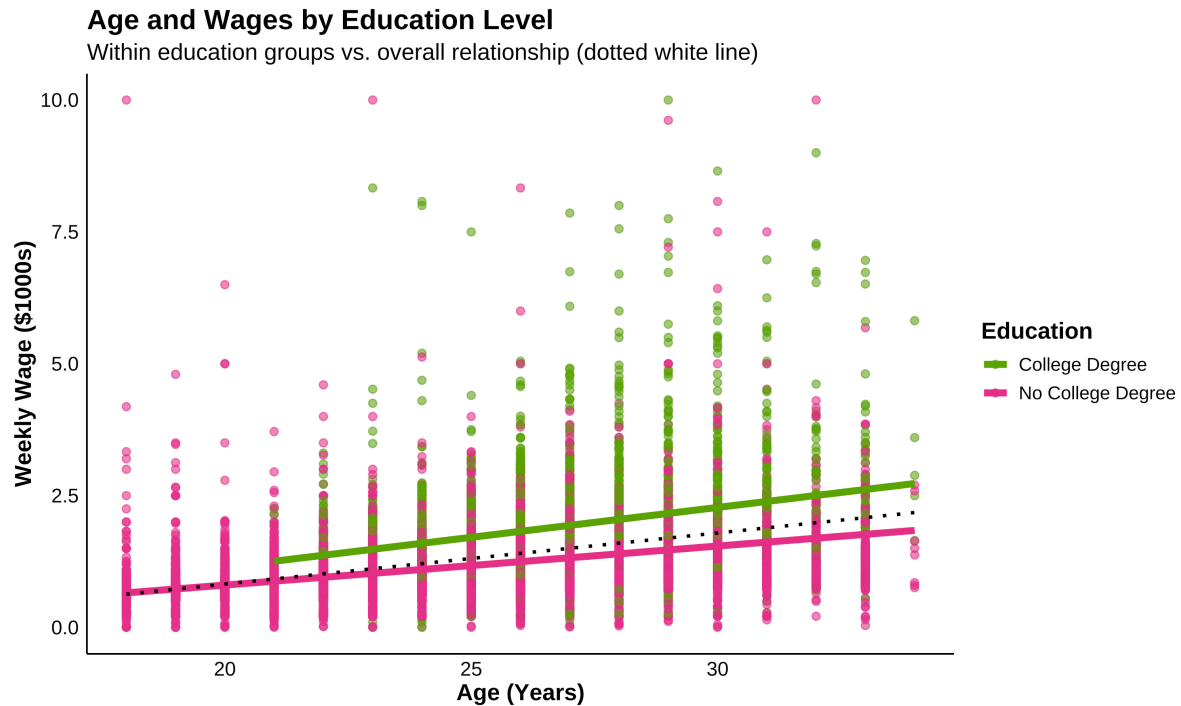
```
# A tibble: 3 x 5
```

	term	estimate	std.error	statistic	p.value
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	(Intercept)	-0.769	0.133	-5.79	7.16e- 9
2	age	0.0780	0.00558	14.0	4.68e-44
3	college_or_more	0.590	0.0568	10.4	3.02e-25

- The age coefficient **changed** once education entered the model
- The education effect dwarfs it: a college degree **7.6 years** of age

Visualizing the Age-Wage Relationship

```
`geom_smooth()` using formula = 'y ~ x'  
`geom_smooth()` using formula = 'y ~ x'
```



From One Slope to Two

Returning to the age + college wage model from a few slides back, here's what the math looks like in general form.

Simple regression (one predictor):

$$\hat{Y} = \alpha + \beta X$$

One intercept, one slope → one line through the data

Multivariate regression (two predictors):

$$\hat{Y} = \alpha + \beta_1 X_1 + \beta_2 X_2$$

One intercept (α), **two slopes** (β_1, β_2) $\rightarrow$ each slope tells you the effect of one variable, holding the other fixed

Parallel Lines: One Slope, Shifted Intercepts

Our wage model: $\widehat{Wage} = \alpha + \beta_1 \cdot Age + \beta_2 \cdot College$

For non-college workers ($College = 0$):

$$\widehat{Wage} = \alpha + \beta_1 \cdot Age$$

For college workers ($College = 1$):

$$\widehat{Wage} = (\alpha + \beta_2) + \beta_1 \cdot Age$$

Same slope (β_1), different intercept — college shifts the line up by β_2

Assumption: Parallel Slopes

Multivariate regression without interactions assumes the effect of **X** on **Y** is identical across every value of the other predictors.

In the wage model `lm(wage ~ age + college)`:

- We are assuming **one extra year of age has the same effect on wages** for college graduates and non-college workers.
- Geometrically: the two lines (college vs. non-college) are **parallel** — same slope, different intercepts.

This is often wrong in real political data. Examples where slopes likely differ:

- The effect of unemployment on approval may be larger for the **incumbent's party** than for the opposition
- The effect of campaign spending may be larger for **challengers** than for incumbents
- The effect of age on turnout may be different for **first-time voters** vs. habitual voters

When you suspect different slopes, fit an **interaction model** ($Y \sim X * Z$) — that's Segment B of today's class.

Interpreting Multivariate Coefficients

Each coefficient tells you the effect of that variable **among observations that are identical on all other variables in the model**

age coefficient: Among people with the **same education level**, each additional year of age is associated with a \$78 increase in wages

college_or_more coefficient: Among people who are the **same age**, college graduates earn an additional \$590 per week

Making a prediction: For a 30-year-old with a college degree:

Predicted wage = $-0.77 + 0.08 \times 30 + 0.59 \times 1 = \text{\$2160}$ per week

Computing Predictions with Controls

The formula: $\hat{Y} = \alpha + \beta_1 X_1 + \beta_2 X_2 + \dots$

Plug in values for **every** variable in the model — both the predictor you care about and the controls.

```
# Our multivariate model: wage ~ age + college_or_more  
tidy(multi_model)
```

```
# A tibble: 3 x 5
```

term <chr>	estimate <dbl>	std.error <dbl>	statistic <dbl>	p.value <dbl>
1 (Intercept)	-0.769	0.133	-5.79	7.16e- 9
2 age	0.0780	0.00558	14.0	4.68e-44
3 college_or_more	0.590	0.0568	10.4	3.02e-25

Prediction Examples: By Hand

$$\hat{Y} = \alpha + \beta_1 \cdot \text{Age} + \beta_2 \cdot \text{College}$$

Person A: 25, no college

$$\hat{Y} = -0.77 + 0.08 \times 25 + 0.59 \times 0$$

$$= -0.77 + 1.95 + 0$$

$$= 1.18 \text{ (thousands)}$$

Predicted wage: **\\$1180/week**

Person B: 25, college degree

$$\hat{Y} = -0.77 + 0.08 \times 25 + 0.59 \times 1$$

$$= -0.77 + 1.95 + 0.59$$

$$= 1.77 \text{ (thousands)}$$

Predicted wage: **\$1771/week**

The difference ($\$1771 - \$1180 = \textbf{\$590}$) is exactly the college coefficient — the predicted effect of college **at the same age**.

Model Building Strategies

How Many Variables Should You Include?

The bias-variance tradeoff:

Bias: Systematic error from wrong assumptions (missing important variables)

- Your estimates consistently miss the true effect
- **Example:** Age appears to strongly predict wages, but it's really education doing the work

Variance: Sensitivity to small changes in your data (too many variables)

- Your estimates jump around with different samples
- **Example:** Including 50 variables makes predictions unstable and unreliable

The Tradeoff:

- **Too few variables:** High bias (omitted variable bias), low variance
- **Too many variables:** Low bias, high variance (overfitting and multicollinearity)

Visualizing the Bias-Variance Tradeoff

Warning in geom_point(aes(x = 0, y = 0), color = "#e6308a", size = 4, inherit.aes = FALSE):
i Please consider using `annotate()` or provide this layer with data containing a single row.



We want models in the top-left quadrant - close to the truth (low bias) and consistent across samples (low variance)

Choosing Control Variables

The key question: Does this variable plausibly affect **both** my main predictor (X) **and** my outcome (Y)?

Include a variable as a control if:

- Theory or prior research identifies it as a confounder
- It is a plausible common cause of X and Y (creates a “backdoor path”)
- Leaving it out would bias your estimate of X’s effect

Don't include a variable if:

- It is a **consequence** of X (a mediator) — controlling for it absorbs the very effect you're trying to measure
- It is unrelated to either X or Y — just adds noise and wastes degrees of freedom
- It is measured **after** the treatment/main predictor (post-treatment bias) — e.g., if education → income → health, controlling for income absorbs part of education's effect on health

Exploring Presidential Approval

```
# Load presidential approval data
approval_data <- read_csv("../data/presidential_approval.csv") %>%
  mutate(
    year = as.numeric(format(as.Date(date), "%Y")),
    approve = approval_rating
  ) %>%
  filter(!is.na(approve), !is.na(year)) %>%
  mutate(
    post_9_11 = ifelse(year >= 2001, 1, 0),
    post_2008_crisis = ifelse(year >= 2008, 1, 0),
    decades_since_1980 = (year - 1980) / 10
  )
```

Rows: 528 Columns: 2

-- Column specification -----

Delimiter: ","

dbl (1): approval_rating

date (1): date

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
# Look at the data structure
```

```
head(approval_data)
```

```
# A tibble: 6 x 7
```

	date	approval_rating	year	approve	post_9_11	post_2008_crisis
	<date>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	1980-01-15	58.2	1980	58.2	0	0

2	1980-02-15	55.8	1980	55.8	0	0
3	1980-03-15	39.4	1980	39.4	0	0
4	1980-04-15	40.1	1980	40.1	0	0
5	1980-05-15	43.2	1980	43.2	0	0
6	1980-06-15	31.8	1980	31.8	0	0

i 1 more variable: decades_since_1980 <dbl>

Building Models Sequentially

```
tidy(lm(approve ~ decades_since_1980, data = approval_data))
```

```
# A tibble: 2 x 5
  term          estimate std.error statistic  p.value
<chr>          <dbl>    <dbl>    <dbl>    <dbl>
1 (Intercept)    57.3      0.921     62.3 8.16e-245
2 decades_since_1980 -3.21    0.369    -8.72 3.76e- 17
```

```
tidy(lm(approve ~ decades_since_1980 + post_9_11, data = approval_data))
```

```
# A tibble: 3 x 5
  term          estimate std.error statistic  p.value
<chr>          <dbl>    <dbl>    <dbl>    <dbl>
1 (Intercept)    56.3      0.995     56.6 1.30e-225
2 decades_since_1980 -1.53    0.732     -2.10 3.64e- 2
3 post_9_11      -4.94      1.86     -2.65 8.22e- 3
```

```
tidy(lm(approve ~ decades_since_1980 + post_9_11 + post_2008_crisis, data = approval_data))
```

```
# A tibble: 4 x 5
  term          estimate std.error statistic  p.value
<chr>          <dbl>    <dbl>    <dbl>    <dbl>
1 (Intercept)    53.6      1.11     48.2 8.63e-195
2 decades_since_1980  1.20    0.895      1.34 1.80e- 1
3 post_9_11      -4.90      1.82     -2.69 7.27e- 3
4 post_2008_crisis -8.70      1.71     -5.09 5.11e- 7
```

Watching Coefficients Move

What to notice: watch the `decades_since_1980` coefficient as each event indicator enters.

- When a coefficient **moves** after adding a control, the two variables were entangled — the simple trend was partly absorbing those events
- When it **holds steady**, the original estimate was robust to that control

This before/after comparison is the everyday habit of multivariate analysis.

An Example: Afghan Survey Data

The Afghan Civilian Survey

The Study: Survey of Afghan civilians during the war (ISAF/Taliban conflict)

Research Question: Which demographic factors predict whether civilians experienced violence by armed groups?

Key Variables:

- **Outcome:** `violence` - measure experiences of ISAF violence (yes or no)
- **Predictors:** Age, education years, employment status
- **Context:** Rural Afghan civilians navigating conflict between international forces and Taliban

Why This Matters:

- Understanding **civilian experiences** in conflict zones
- Examining how **socioeconomic factors** predict exposure to wartime violence
- Testing whether **age and education effects** change when controlling for employment

Multiple Predictors of Violence Experiences

```
# Load and explore Afghan survey data
afghan_data <- read_csv("../data/afghan.csv") %>%
  filter(!is.na(violent.exp.ISAF), !is.na(age), !is.na(educ.years), !is.na(income)) %>%
  # Create cleaner variable names
  rename(
    violence = violent.exp.ISAF,
```

```
education = educ.years
)
```

Rows: 2754 Columns: 11

-- Column specification -----

Delimiter: ","

chr (4): province, district, income, list.group

dbl (7): village.id, age, educ.years, employed, violent.exp.ISAF, violent.ex...

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
# Look at the data
```

```
head(afghan_data) %>%
```

```
  select(age, education, employed, violence)
```

```
# A tibble: 6 x 4
```

	age	education	employed	violence
	<dbl>	<dbl>	<dbl>	<dbl>
1	26	10	0	0
2	49	3	1	0
3	60	0	1	1
4	34	14	1	0
5	21	12	1	0
6	42	6	1	0

Four Models: Comparing Different Predictors

```
# Model 1: Just age
```

```
model1 <- lm(violence ~ age, data = afghan_data)
```

```
# Model 2: Just employed
```

```
model2 <- lm(violence ~ employed, data = afghan_data)
```

```
# Model 3: Just education
```

```
model3 <- lm(violence ~ education, data = afghan_data)
```

```
# Model 4: All three predictors
```

```
model4 <- lm(violence ~ age + employed + education, data = afghan_data)
```


Model Comparison Table

Warning: package 'knitr' was built under R version 4.5.2

Table 5: Coefficient Estimates: How Effects Change Across Models

Variable	Age Only	Employed Only	Education Only	Full Model
(Intercept)	0.37	0.401	0.442	0.442
Age	0	—	—	0.001
Employed	—	-0.044	—	-0.035
Education (years)	—	—	-0.016	-0.016
N	2584	2584	2584	2584

- How do coefficients change when adding other variables?
- Which individual predictor has the largest effect?
- Do the signs and magnitudes make substantive sense?

Understanding Model Fit: R^2

R^2 with Multiple Predictors

Recall from Week 5: R^2 is the share of variance in Y your model explains (0 to 1), and whether a value is “good” depends on context.

What’s new with multiple predictors:

R^2 **never goes down when you add a variable** — even a useless one

- OLS can set the new coefficient to ~ 0 , so fit can’t get worse
- Random noise usually nudges R^2 *up*

Adjusted R^2 charges a small penalty per predictor

- Rises only when a variable explains **more than chance would**
- *Falls* when you add junk → use it to compare models

R^2 for Our Afghan Violence Models

Table 6: Model Fit: How Much Variation Do We Explain?

Model	R^2	Adjusted R^2	% Variance Explained
Age Only	0.0000	-0.0004	0%
Employed Only	0.0020	0.0016	0.2%
Education Only	0.0260	0.0257	2.6%
Full Model	0.0274	0.0263	2.7%

Questions to Consider:

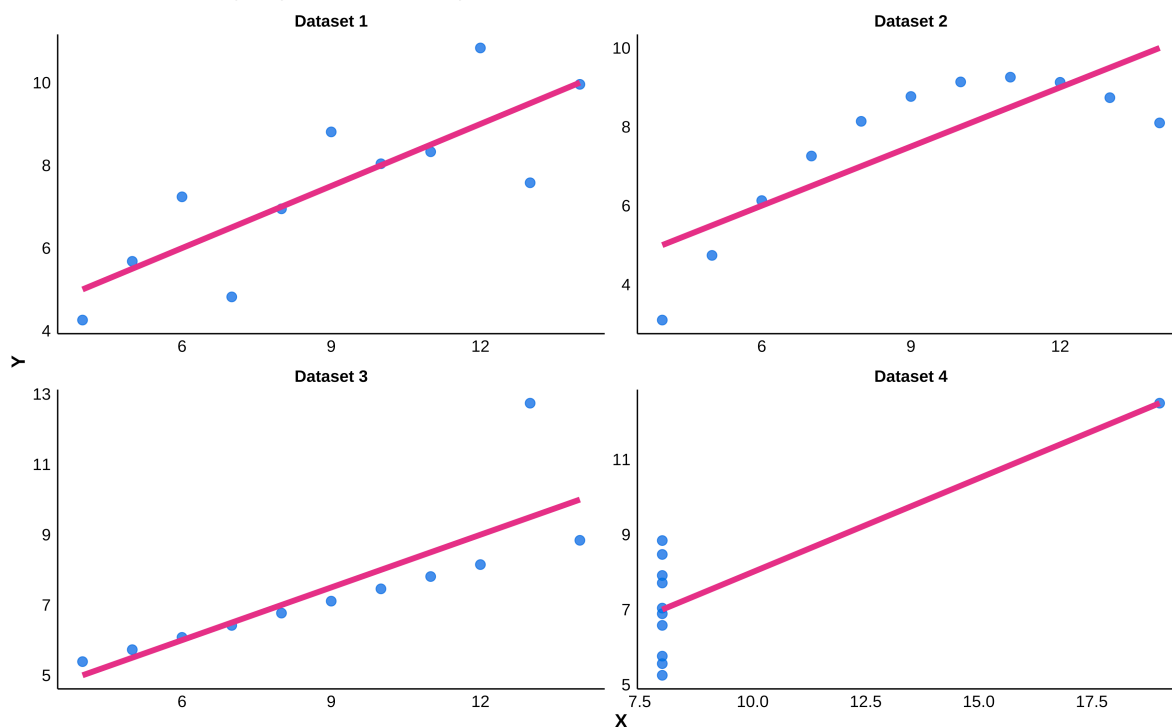
- Which single variable is most predictive?
- The full model has the highest R^2 — it must, since it contains every predictor. Does its **adjusted R^2** say the extra variables earned their place?
- Are these values “good”? (Context matters — recall Week 5)

Anscombe's Quartet: Why R^2 Isn't Everything

``geom_smooth()`` using formula = 'y ~ x'

Anscombe's Quartet: All Have $R^2 = 0.667$

Same statistics, completely different relationships!



Lesson: Always visualize your data! Statistics alone can be misleading.

The Danger of R^2 Hunting

R^2 Hunting: Keep adding variables until you get a high R^2

Problem: This leads to **overfitting**

What is Overfitting?

- Model fits your specific dataset very well
- But performs poorly on new data
- Like memorizing answers instead of understanding concepts

Solution: Have **principled reasons** for including variables

- Theory suggests they're important confounders
- Prior research shows they matter
- They're correlated with both X and Y

Remember: A model with $R^2 = 0.3$ that makes theoretical sense is better than $R^2 = 0.8$ with random variables

Cautions and Practical Guidelines

What Multivariate Regression Can and Cannot Do

What it CAN do:

- Control for **observed** confounders
- Estimate each variable's partial effect
- Test whether a relationship survives controls

What it CANNOT do:

- Control for **unobserved** confounders
- Prove causation
- Fix a poor research design

Three pitfalls to avoid:

- **Overfitting:** too many variables relative to sample size — R^2 climbs, but the model is memorizing noise
- **Multicollinearity:** highly correlated predictors make estimates unstable
- **Post-treatment bias:** controlling for a *consequence* of your main predictor absorbs the very effect you want to measure

Theory should guide which variables to include, not just statistical fit — and be honest about what your model can't rule out.

Segment B — Categorical Predictors and Interactions

Putting Categories into a Regression

Problem: Regression needs numbers, but politics is full of categories

Examples:

- Party affiliation (Republican, Democrat, Independent, ...)
- Nationality (American, Canadian, British, ...)
- Gender (Male, Female, ...)
- Education level (High School, College Graduate, ...)

Solution: Use categorical variables (factors) directly in regression models

Factors in R

Factor: R's data type for categorical variables

- Stores the category labels (“White”, “Black”, “Hispanic”, “Asian”) plus an internal numeric coding
- In regression, **R automatically expands a factor into 0/1 dummy variables** — one per category, minus a reference (next slide)
- Order matters only for **ordinal** variables, not **nominal** ones:
 - Education is ordinal — “college graduate” *is* more than “high school graduate”
 - Race is nominal — “Asian” is not more than “White”

How Categorical Variables Become Numbers

We need to be able to represent categorical variables in a model. We will “decompose” the categorical variable into multiple dummy variables.

The decomposition process: One categorical variable → Multiple dummy variables

- For example, if we have a variable **race** with four levels: “White”, “Black”, “Hispanic”, “Asian”, we can create three dummy variables: **Black**, **Hispanic**, and **Asian**.
- The reference category is “White”.
- The dummy variables are binary variables that are 1 if the observation is in the level and 0 otherwise.
- For example, if the observation is “Black”, then **Black** is 1 and the other dummy variables are 0.

```
# A tibble: 4 x 5
  respondent_id race      Black Hispanic Asian
      <int> <chr>      <dbl>      <dbl> <dbl>
1           1 White         0          0      0
2           2 Black         1          0      0
3           3 Hispanic       0          1      0
4           4 Asian          0          0      1
```

Party and Vote Share in R

```
Rows: 2874 Columns: 44
-- Column specification -----
Delimiter: ","
chr (8): bonica.rid, election, Cand.ID, Name, State, seat, District, cand.g...
dbl (36): cycle, Party, cfscore, male.winner, running.variable, female.margi...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
# Model with party factor - R creates dummy variables automatically
party_model <- lm(vote_share ~ party, data = campaign_data)
tidy(party_model)
```

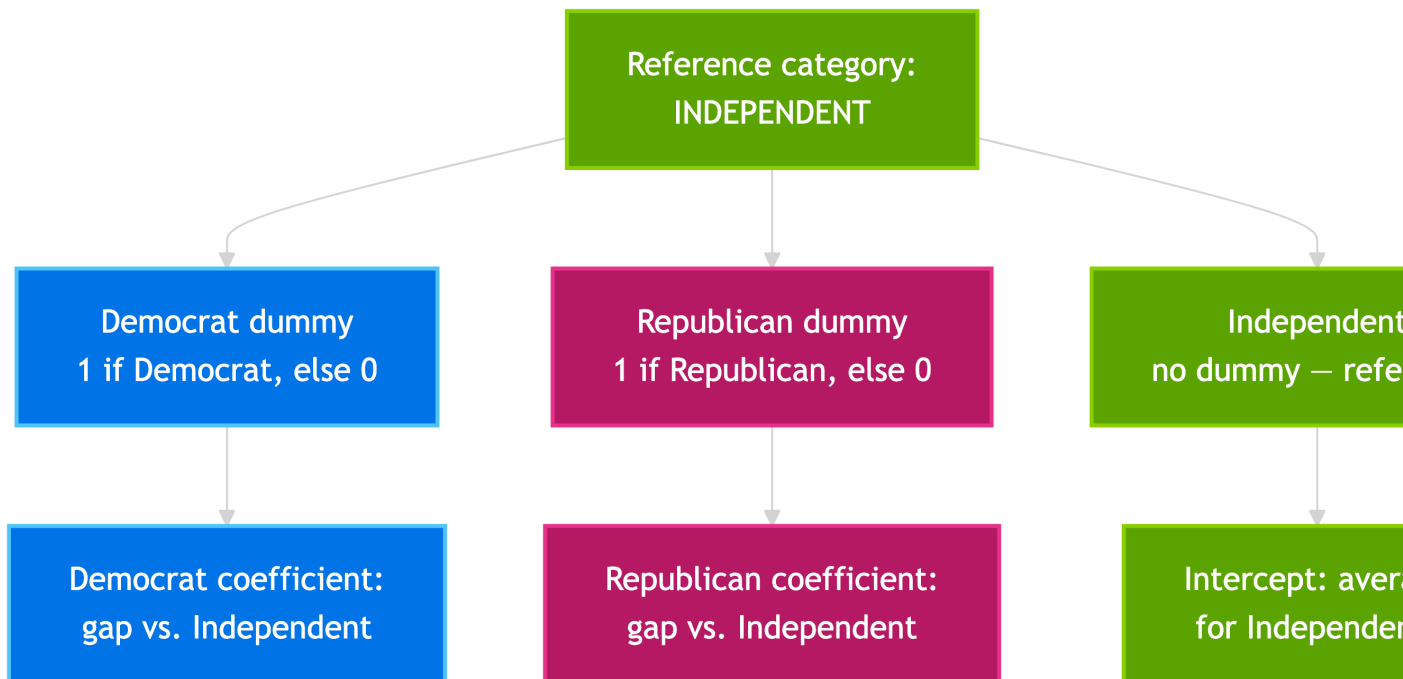
```
# A tibble: 3 x 5
  term          estimate std.error statistic p.value
  <chr>          <dbl>      <dbl>      <dbl>    <dbl>
1 (Intercept)    13.1        21.4      0.611    0.542
```

2	partyRepublican	14.9	21.5	0.695	0.487
3	partyDemocrat	12.1	21.5	0.563	0.574

Interpretation:

- **Intercept:** Average vote share for reference category (Independent)
- **partyRepublican:** How much more Republicans get compared to Independents
- **partyDemocrat:** How much more Democrats get compared to Independents

The Reference Category



Every coefficient tells you the difference from the reference group.

Why Do We Need a Reference?

Suppose we included all k dummies — Democrat, Republican, *and* Independent:

- The three dummies would always sum to 1 for every observation — exactly the same as the model's intercept
- That's perfect multicollinearity (the “**dummy variable trap**”): the model can't tell the intercept apart from the dummies

- R will silently drop one dummy for you if you try

Dropping one category ourselves makes the comparison explicit: each coefficient is the gap from the dropped (reference) category.

Adding Multiple Categorical Predictors

```
# Model with party and gender factors
multi_categorical_model <- lm(vote_share ~ party + gender_factor, data = demo_campaign)
tidy(multi_categorical_model)
```

A tibble: 4 x 5

term	estimate	std.error	statistic	p.value
<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1 (Intercept)	48909.	1896.	25.8	0.0000134
2 partyRepublican	-4030.	2190.	-1.84	0.140
3 partyDemocrat	4697.	2190.	2.14	0.0985
4 gender_factorFemale	-818.	1756.	-0.466	0.665

Interpretation:

- **partyRepublican**: Republicans vs Others, holding gender constant
- **partyDemocrat**: Democrats vs Others, holding gender constant
- **genderFemale**: Women vs Men, holding party constant

More Dummy-Variable Examples

Table 7: Categorical Variable Decomposition Examples

Variable	All Levels	Reference Level	Dummy Variables Estimated
Gender	Male, Female	Male	Female
Education Level	High School, College, Graduate	High School	College, Graduate
Geographic Region	Northeast, South, Midwest, West	Northeast	South, Midwest, West
Vote Choice	Democrat, Republican, Independent	Independent	Democrat, Republican

Variable	All Levels	Reference Level	Dummy Variables Estimated
Income Level	Low, Middle, High, Very High	Low	Middle, High, Very High

Pattern: For K categories, we estimate K-1 dummy variables. The reference category is excluded to avoid perfect collinearity.

Interaction Terms: When Relationships Depend on Context

Interactions

Interaction effect: How the relationship between X and Y changes depending on a third variable Z

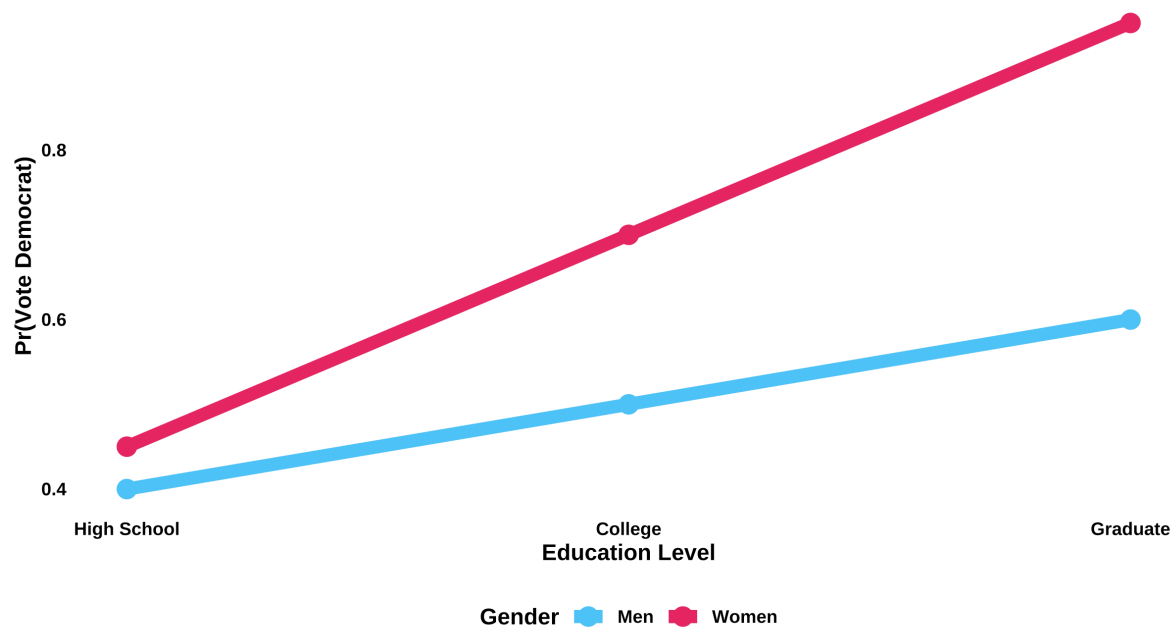
Political examples:

- Does education affect voting differently for men vs women?
- Does economic performance matter more for incumbent vs challenger candidates?
- Do campaign ads work differently in competitive vs safe districts?

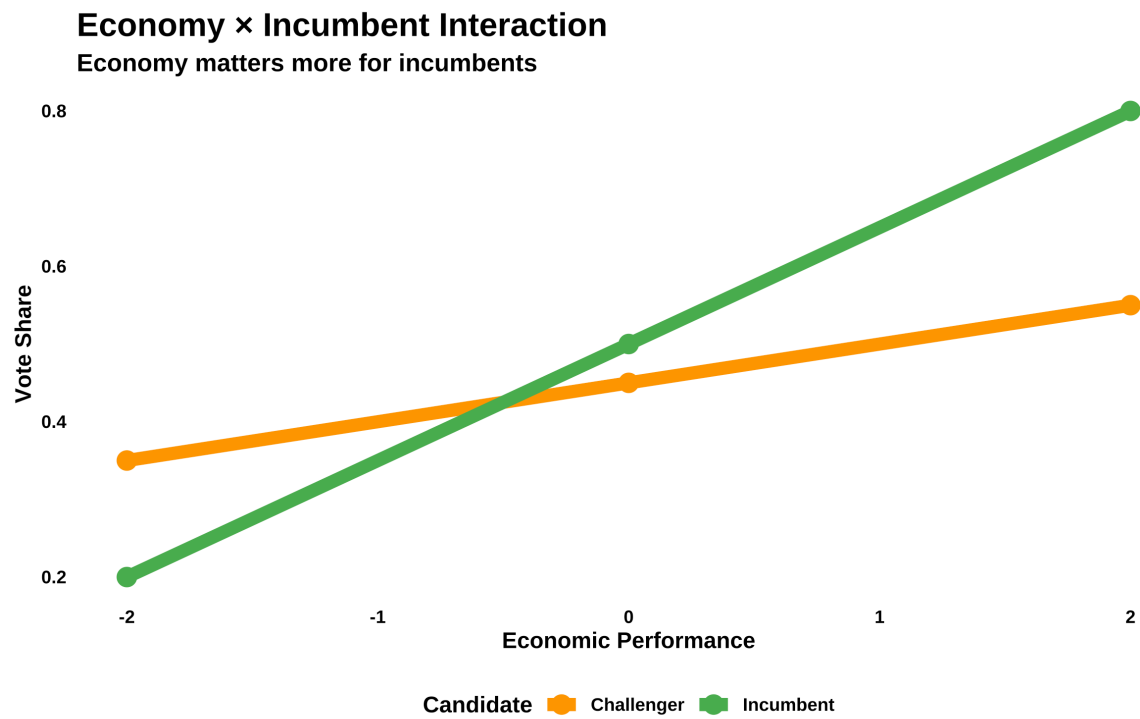
Example 1: Education × Gender Interaction

Education × Gender Interaction

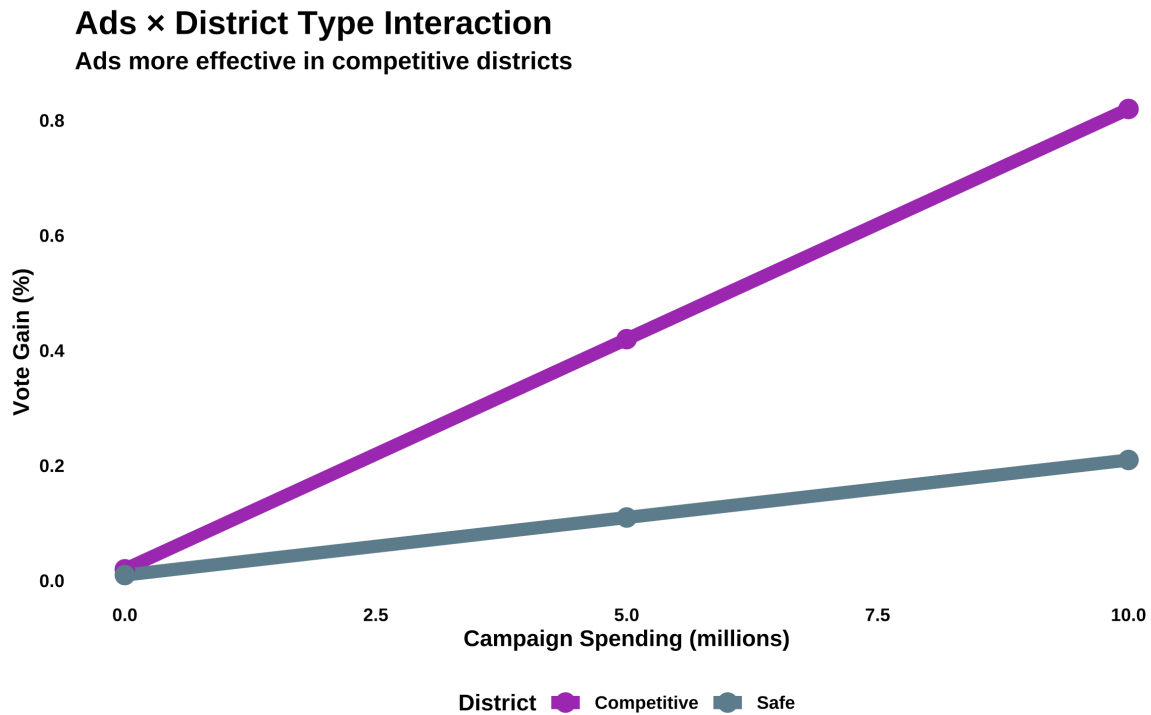
Education effect stronger for women



Example 2: Economy × Incumbent Interaction

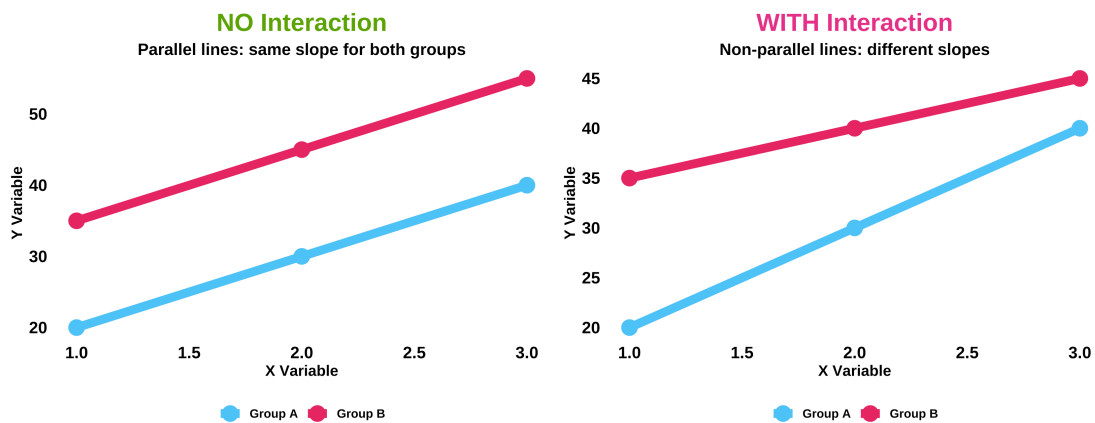


Example 3: Campaign Ads × District Type Interaction



Notice the different slopes** for each group! When lines are parallel = no interaction. When slopes differ = interaction present.

Visualizing Interactions: Parallel vs. Non-Parallel Lines



Key insight: Look at the slopes! If lines are parallel, there's no interaction. If lines cross or diverge, there IS an interaction - the effect of X depends on which group you're in.

What Is an Interaction Term?

Interaction Term: An interaction term is the product $X * Z$. Its coefficient says **how much X's slope changes as Z changes** — i.e., the effect of one variable **depends on** the level of another.

Think of it this way:

- **Main effect:** “Does X affect Y?”
- **Interaction effect:** “Does X affect Y **differently** for different groups?”

Example: Do Republicans and Democrats justify political violence differently based on shooter's party?

- Main effect: Party ID $\rightarrow$ Violence justification
- Interaction: Party ID $\times$ Shooter alignment $\rightarrow$ Justification varies by alignment

Mathematical representation:

$$Y = \beta_0 + \beta_1 X + \beta_2 Z + \beta_3 (X \times Z) + \epsilon$$

Where β_3 captures how the effect of X on Y changes as Z changes.

Creating Interaction Terms

To create an interaction term, you can use the `*` operator in the formula.

```
Rows: 1863 Columns: 7
```

```
-- Column specification -----
```

```
Delimiter: ","
```

```
chr (4): km_support, pid, passed, alignment
```

```
dbl (3): supportactions, justified, charged
```

```
i Use `spec()` to retrieve the full column specification for this data.
```

```
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
# Model with interaction between party ID and shooter alignment
interaction_model <- lm(violence_justified ~ party_id*shooter_alignment,
                        data = violence_data)

tidy(interaction_model)
```

A tibble: 4 x 5

term <chr>	estimate <dbl>	std.error <dbl>	statistic <dbl>	p.value <dbl>
1 (Intercept)	0.0979	0.0131	7.45	1.42e-13
2 party_idRepublican	-0.00883	0.0189	-0.467	6.41e- 1
3 shooter_alignmentIn-Party Shooter	0.00886	0.0185	0.478	6.32e- 1
4 party_idRepublican:shooter_alignmentIn-~	-0.0286	0.0267	-1.07	2.84e- 1

- **party_idRepublican** (β_1): the Republican–Democrat difference **when the shooter is out-party** (the reference condition)
- **shooter_alignmentIn-Party Shooter** (β_2): the in-party vs. out-party difference **among Democrats** (the reference party)
- **party_idRepublican:shooter_alignmentIn-Party Shooter** (β_3): how much the in-party/out-party difference **changes** for Republicans compared to Democrats

“Main effects” here are **conditional** — each is the effect when the other variable is at its reference (0) level, not an overall average.

The Four Predicted Values

With two binary variables, the model defines **four predicted values**. Build each one from the coefficients:

Table 8: Predicted Justification of Violence by Group

Group	Built From	Predicted
Democrat, Out-Party Shooter	Intercept	0.098
Republican, Out-Party Shooter	Intercept + 1	0.089
Democrat, In-Party Shooter	Intercept + 2	0.107
Republican, In-Party Shooter	Intercept + 1 + 2 + 3	0.069

β_3 is the “**difference in differences**” — how much the in-party/out-party gap differs between the two parties. If $\beta_3 = 0$, both parties react identically.

Interaction with Continuous Variables

Rows: 1863 Columns: 7

-- Column specification -----

Delimiter: ","

chr (4): km_support, pid, passed, alignment

dbl (3): supportactions, justified, charged

i Use ``spec()`` to retrieve the full column specification for this data.

i Specify the column types or set ``show_col_types = FALSE`` to quiet this message.

```
# Model with interaction between two continuous variables
continuous_interaction <- lm(violence_justified ~ age * political_engagement, data = continuu
tidy(continuous_interaction)
```

A tibble: 4 x 5

	term	estimate	std.error	statistic	p.value
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	(Intercept)	0.0872	0.0698	1.25	0.212
2	age	0.0000790	0.00145	0.0544	0.957
3	political_engagement	-0.000104	0.00130	-0.0799	0.936
4	age:political_engagement	0.00000209	0.0000271	0.0773	0.938

Interpreting the interaction coefficient (here 0):

- It is how much the **age slope changes** per one-point increase in engagement (symmetric: also the reverse)
- Here it's essentially **zero** — the age slope is the same at every engagement level
- **A null interaction is a real finding:** the simpler additive model was adequate

Unlike categorical interactions where we compare groups, continuous interactions tell us how **slopes change** as one variable increases.

When to Use Interactions

Good and Bad Reasons for an Interaction

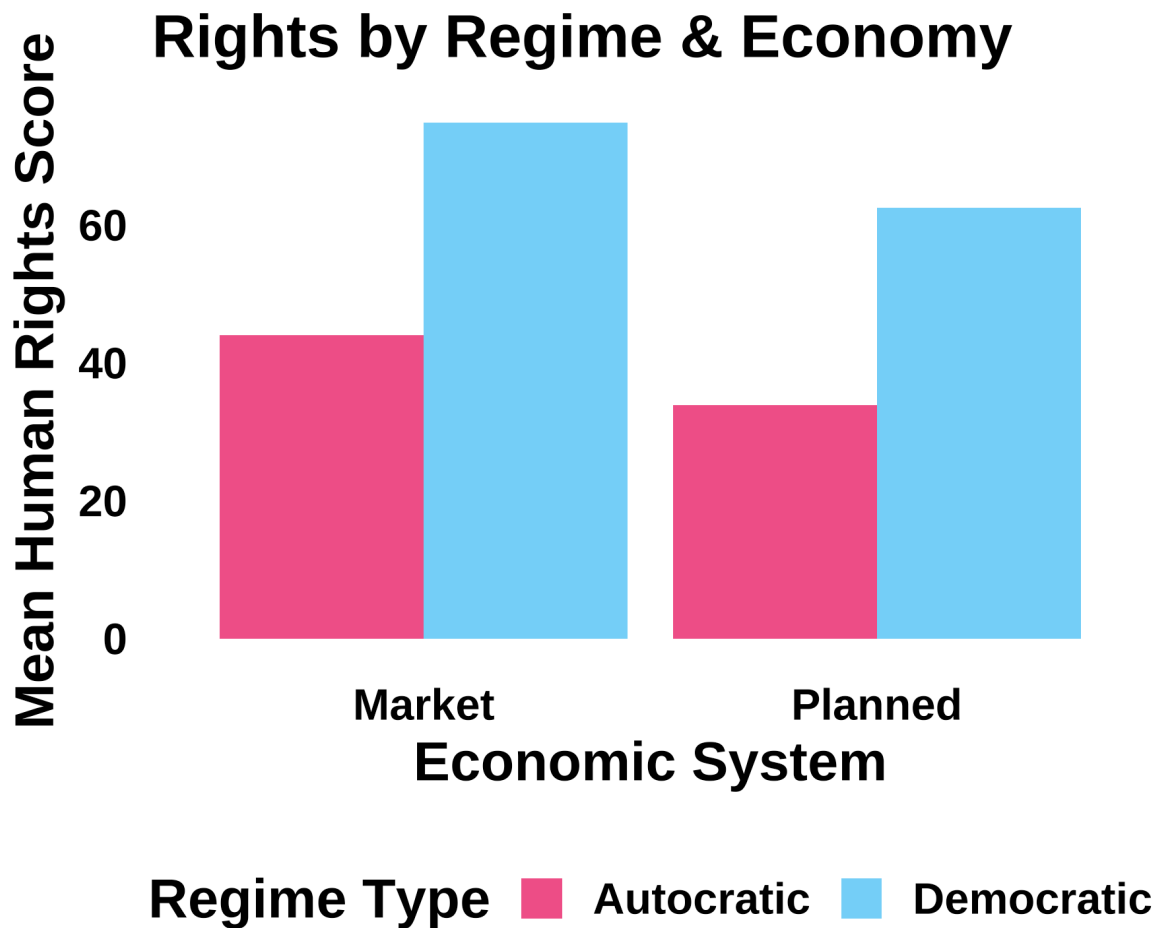
Good reasons for interactions:

- Theory predicts conditional relationships
- Different groups should respond differently
- Previous research suggests moderation effects
- Substantively important to test

Poor reasons for interactions:

- Fishing for significance
- Adding complexity without theory

Example 1: Regime Type × Economic System (Factor × Factor)



```
# Regression model
regime_model <- lm(human_rights ~ regime_type * economic_system,
  data = regime_data)
```

```
tidy(regime_model)
```

```
# A tibble: 4 x 5
  term                                estimate std.error statistic  p.value
  <chr>                                <dbl>    <dbl>    <dbl>    <dbl>
1 (Intercept)                        44.0      1.97     22.4 6.52e-40
2 regime_typeDemocratic                30.8      2.78     11.1 6.72e-19
```


3	economic_systemPlanned	-10.1	2.84	-3.55	6.05e- 4
4	regime_typeDemocratic:economic_systemPl~	-2.25	3.94	-0.571	5.69e- 1

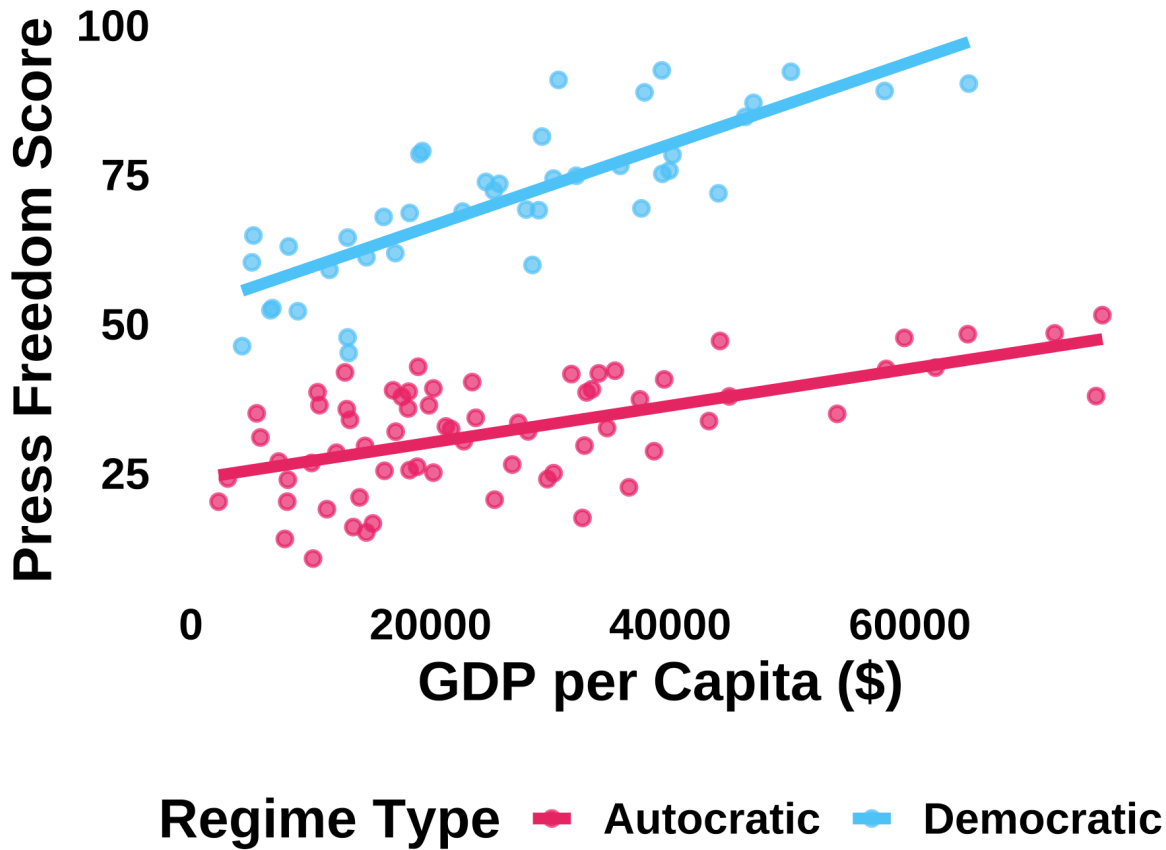
```
# Extract coefficients for inline use
regime_coefs <- coef(regime_model)
regime_dem <- round(regime_coefs["regime_typeDemocratic"], 1)
regime_planned <- round(regime_coefs["economic_systemPlanned"], 1)
regime_interact <- round(regime_coefs["regime_typeDemocratic:economic_systemPlanned"], 2)
regime_dem_planned_effect <- round(abs(regime_planned) + abs(regime_interact), 1)
```

- Democracies have 30.8 points higher human rights scores than autocracies (with market economies)
- Planned economies reduce human rights by 10.1 points in autocracies
- The interaction (-2.25) shows planned economies reduce human rights slightly more in democracies, but this effect is not statistically significant
- Overall: Market economies boost human rights by 10.1 points in autocracies vs 12.3 points in democracies

Example 2: GDP per Capita × Regime Type (Continuous × Factor)

```
`geom_smooth()` using formula = 'y ~ x'
```

Press Freedom by GDP & Regime



```
# Regression model
gdp_model <- lm(press_freedom ~ gdp_per_capita * regime_type,
                data = gdp_data)

tidy(gdp_model)
```

```
# A tibble: 4 x 5
  term                                estimate std.error statistic  p.value
  <chr>                                <dbl>    <dbl>    <dbl>    <dbl>
1 (Intercept)                        24.0      1.66     14.5 9.59e-27
2 gdp_per_capita                      0.000309 0.0000521  5.93 3.90e- 8
3 regime_typeDemocratic                28.6      2.92     9.80 1.73e-16
4 gdp_per_capita:regime_typeDemocratic 0.000378 0.0000945  4.00 1.18e- 4
```

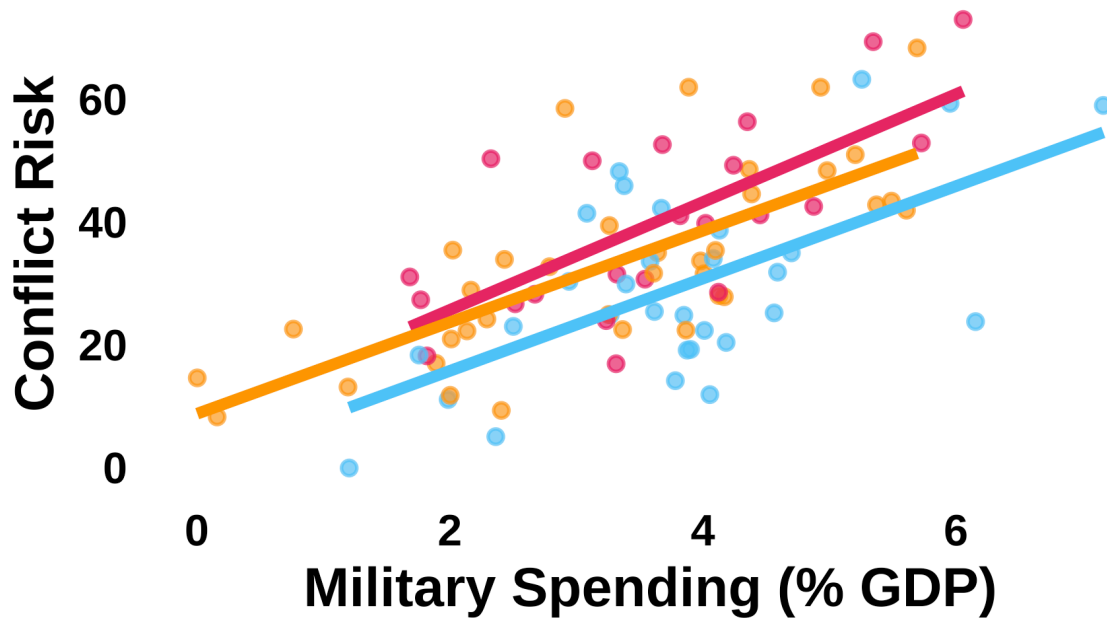
```
# Extract coefficients for inline use (multiply by 1000 for per-$1000 interpretation)
gdp_coefs <- coef(gdp_model)
gdp_autocracy <- round(gdp_coefs["gdp_per_capita"] * 1000, 3)
gdp_dem_base <- round(gdp_coefs["regime_typeDemocratic"], 1)
gdp_interact <- round(gdp_coefs["gdp_per_capita:regime_typeDemocratic"] * 1000, 3)
gdp_democracy <- round(gdp_autocracy + gdp_interact, 3)
```

- In autocracies: Each \$1,000 increase in GDP per capita increases press freedom by 0.309 points
- In democracies: Each \$1,000 increase in GDP per capita increases press freedom by 0.687 points (0.309 + 0.378)
- The interaction shows that GDP has more than twice the effect on press freedom in democracies compared to autocracies
- Democracies start with 28.6 points higher press freedom than autocracies at the same GDP level

Example 3: Military Spending × Trade Openness (Continuous × Continuous)

```
`geom_smooth()` using formula = 'y ~ x'
```

Conflict Risk: Spending & Trade



Trade Openness

Low Medium High

```
# Regression model
conflict_model <- lm(conflict_risk ~ military_spending * trade_openness,
                     data = conflict_data)

tidy(conflict_model)
```

```
# A tibble: 4 x 5
  term                estimate std.error statistic  p.value
  <chr>                <dbl>    <dbl>    <dbl>    <dbl>
1 (Intercept)         19.4      11.6      1.67  0.0981
2 military_spending    10.1       2.93     3.43 0.000935
3 trade_openness      -0.210     0.184    -1.14  0.257
4 military_spending:trade_openness -0.0384  0.0459   -0.836 0.405
```

Interpretation: Military spending increases conflict risk more when trade openness is low. Higher trade integration dampens the conflict-inducing effects of military buildup.

Making Predictions from Interaction Models

Example 1: Education Policy - Class Size × Teacher Experience

```
edu_model <- lm(test_scores ~ class_size * teacher_experience + school_funding, data = education_data)
tidy(edu_model)
```

```
# A tibble: 5 x 5
  term                estimate std.error statistic  p.value
  <chr>                <dbl>    <dbl>    <dbl>    <dbl>
1 (Intercept)         75.4      4.43      17.0 3.06e-39
2 class_size          -1.07     0.164     -6.48 8.53e-10
3 teacher_experience    0.416    0.389      1.07 2.87e- 1
4 school_funding       0.000133 0.0000222    6.02 9.91e- 9
5 class_size:teacher_experience 0.0304 0.0152      2.00 4.67e- 2
```

Prediction Question: What are the expected test scores for a class of 20 students with a teacher who has 5 years of experience, in a school with \$45,000 funding?

Manual Calculation:

$$\begin{aligned}\hat{Y} &= 75.4 + (-1.07 \times 20) + (0.42 \times 5) + (0.00013 \times 45000) + (0.03 \times 20 \times 5) \\ &= 75.4 + -21.3 + 2.1 + 6 + 3.04 = 65.2\end{aligned}$$

Example 2: Political Behavior - Campaign Spending × Incumbent Status

```
campaign_model <- lm(vote_share ~ spending_thousands * incumbent + district_competitiveness, data = campaign_data)
tidy(campaign_model)
```

```
# A tibble: 5 x 5
  term                estimate std.error statistic  p.value
  <chr>                <dbl>    <dbl>    <dbl>    <dbl>
1 (Intercept)         49.7      1.30      38.2 2.80e-55
2 spending_thousands  0.0614  0.00576    10.7 2.46e-17
3 incumbent           4.07     1.97       2.07 4.16e- 2
4 district_competitiveness 0.158  0.0180     8.74 1.77e-13
5 spending_thousands:incumbent -0.0140 0.00987    -1.42 1.59e- 1
```

Prediction Question: What is the expected vote share for a challenger who spends \$300,000 in a district with competitiveness score of 55?

$$\begin{aligned}\hat{Y} &= 49.7 + (0.061 \times 300) + (4.07 \times 0) + (0.158 \times 55) + (-0.014 \times 300 \times 0) \\ &= 49.7 + 18.4 + 0 + 8.7 + 0 = 76.8\end{aligned}$$

Example 3: International Relations - Trade × Regime Type

```
ir_model <- lm(diplomatic_score ~ trade_volume * democracy_score + gdp_per_capita, data = ir,
tidy(ir_model))
```

```
# A tibble: 5 x 5
  term                estimate std.error statistic  p.value
  <chr>                <dbl>    <dbl>    <dbl>    <dbl>
1 (Intercept)         34.0      5.09      6.67 3.45e- 9
2 trade_volume         0.999     0.155     6.43 9.77e- 9
3 democracy_score      0.401     0.0960     4.17 7.86e- 5
4 gdp_per_capita       0.000359 0.0000490     7.31 2.10e-10
5 trade_volume:democracy_score 0.0161  0.00301     5.35 8.63e- 7
```

Prediction Question: What are expected diplomatic relations between countries with \$40 billion trade, democracy score of 70, and GDP per capita of \$30,000?

$$\begin{aligned}\hat{Y} &= 34 + (1 \times 40) + (0.401 \times 70) + (0.000359 \times 30000) + (0.0161 \times 40 \times 70) \\ &= 34 + 39.9 + 28 + 10.8 + 45.1 = 157.8\end{aligned}$$

Example 4: Economics - Minimum Wage × Local Unemployment

```
econ_model <- lm(employment_change ~ min_wage * unemployment_rate + cost_of_living, data = e)
tidy(econ_model)
```

A tibble: 5 x 5

term	estimate	std.error	statistic	p.value
<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1 (Intercept)	-3.83	6.49	-0.590	0.557
2 min_wage	-0.676	0.545	-1.24	0.219
3 unemployment_rate	-1.17	0.941	-1.25	0.217
4 cost_of_living	0.0218	0.0123	1.77	0.0816
5 min_wage:unemployment_rate	0.143	0.0785	1.82	0.0725

Prediction Question: What is the expected employment change for a minimum wage of \$15 in an area with 8% unemployment and cost of living index of 110?

$$\begin{aligned}\hat{Y} &= -3.83 + (-0.676 \times 15) + (-1.17 \times 8) + (0.022 \times 110) + (0.143 \times 15 \times 8) \\ &= -3.83 + -10.14 + -9.38 + 2.39 + 17.19 = -3.76\end{aligned}$$

Limitations and Cautions

The Costs of Extra Interactions

Too many interactions:

- Models become difficult to interpret
- Overfitting risks increase
- Theoretical justification becomes weak

Small sample sizes within groups:

- Interactions require sufficient data in each combination
- Consider collapsing categories if groups are too small

Interpretation complexity:

- Main effects change meaning in presence of interactions
- Conditional effects must be calculated carefully

Controls vs. Interactions: What's the Difference?

Same Ingredients, Different Questions

Both controls and interactions involve adding a third variable Z to a model with X and Y :

- Both involve adding Z alongside X in a regression
- Both require you to think about **why Z matters**
- Both produce models with multiple coefficients

But they answer **fundamentally different questions**:

Control: “Does X affect Y , **accounting for Z ?**” — Z shifts the baseline, but X 's effect is the **same for everyone**

Interaction: “Does X affect Y **differently depending on Z ?**” — X 's effect **changes** based on Z

The Equations Side by Side

Control (Additive) Model

$$Y = \beta_0 + \beta_1 X + \beta_2 Z + \epsilon$$

- β_1 = effect of X on Y (**same** for all values of Z)
- β_2 = how much Z shifts the baseline
- **One slope for X , shifted intercepts**
- R syntax: $Y \sim X + Z$

Interaction Model

$$Y = \beta_0 + \beta_1 X + \beta_2 Z + \beta_3 (X \times Z) + \epsilon$$

- β_1 = effect of X on Y **when $Z = 0$**
- β_3 = how much X 's effect **changes** as Z changes
- **Different slopes for X across values of Z**
- R syntax: $Y \sim X * Z$

The only difference is one term: $\beta_3(X \times Z)$. But that one term changes the entire meaning of the model.

Starting Lines vs. Terrain

Control = Runners on different starting lines

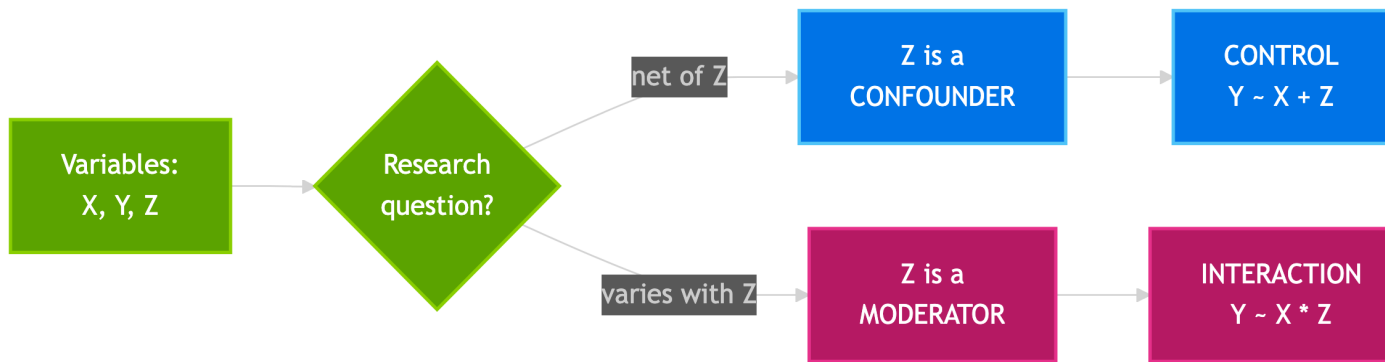
- Everyone runs at the **same speed** (same slope)
- But some start further ahead (different intercepts)
- The race track is **the same** for everyone

Interaction = Runners on different terrains

- Some run on pavement, others on sand
- The terrain **changes how fast** they can run (different slopes)
- The course itself **differs** by group

Controls adjust for **where groups start**. **Interactions** allow for **how groups respond**.

When Should You Use a Control vs. an Interaction?



The key question: Does Z just **shift the baseline** (confounder → control), or does Z **change the relationship** (moderator → interaction)?

Practice: Control or Interaction?

Research Question	Z Variable	Answer	Why?
Does voter turnout affect policy outcomes, accounting for district wealth?	District wealth	Control	Wealth is a confounder — wealthy districts have both higher turnout and different policies
Do sanctions reduce aggression differently for democracies vs. autocracies?	Regime type	Interaction	Asking whether the effect of sanctions depends on regime type
Does social media use predict polarization, controlling for age?	Age	Control	Age affects both social media use and polarization
Does economic anxiety increase populism support more in high-inequality countries?	Inequality	Interaction	Asking whether inequality amplifies the anxiety-populism link

Look for the keywords: “accounting for,” “controlling for,” “holding constant” → **Control**. “Differently for,” “depends on,” “more in,” “moderated by” → **Interaction**.

Common Mistakes with Controls and Interactions

1. Using a control when you need an interaction

- $Y \sim X + Z$ reports one “average” slope (say, 3.5) even if the real effect is **6 for one group and 1 for the other**
- **Detect it:** refit with $Y \sim X * Z$ and check the interaction term, or plot Y vs. X separately by Z and compare slopes

2. Using an interaction when you just need a control

- If Z is a confounder (not a moderator), the interaction adds complexity for nothing — keep it simple when theory doesn’t predict conditional effects

3. Forgetting that interactions include controls

- In R, $Y \sim X * Z$ automatically expands to $Y \sim X + Z + X:Z$ — the interaction model **nests** the control model, so you get the level shift too

Controls vs. Interactions: Summary

	Control	Interaction
Question	Does X affect Y, accounting for Z?	Does X affect Y differently by Z?
Z's role	Confounder (nuisance)	Moderator (substantive)
Equation	$Y = \beta_0 + \beta_1 X + \beta_2 Z$	$Y = \beta_0 + \beta_1 X + \beta_2 Z + \beta_3 XZ$
Lines	Parallel (same slope)	Non-parallel (different slopes)
Coefficients	β_1 = effect of X (for all Z)	β_1 = effect when $Z=0$; β_3 = change per unit Z
Visual	Lines shift up/down	Lines fan out or cross
R syntax	$Y \sim X + Z$	$Y \sim X * Z$

One-line rule: Controls ask “what is X’s effect?” Interactions ask “does X’s effect *vary* with Z?” Both are valid — the choice depends on your research question and theory.

Looking Ahead

In Our Next Class

Sampling and Statistical Significance

- Populations vs. samples, and why sampling creates uncertainty
- Randomization revisited: how chance builds balanced groups
- Margin of error and a first look at the t-distribution
- Hypothesis testing and p-values